Indices - Multiplying Indices (Sheet 1)

Rewrite in expanded form.

Example $5^4 = 5 \times 5 \times 5 \times 5$

1 $3^5 = \square \times \square \times \square \times \square \times \square$

2 $15^3 =$ ______

3 $9^4 =$ ______

4 $1^6 =$ ______

Now algebraic indices - same process

5 $y^4 = \square \times \square \times \square \times \square$

6 $a^5 =$ ______

7 $k^6 =$ ______

8 $b^2 =$ ______

Use expansion to answer these in index form. No calculator.

Example $5^2 \times 5^4 = \overbrace{5 \times 5}^{5^2} \times \overbrace{5 \times 5 \times 5 \times 5}^{5^4} = 5^6$

9 $7^2 \times 7^2 = \square \times \square \times \square \times \square$

$7^2 \times 7^2 = \square^{\square}$

10 $3^3 \times 3^2 =$ ______

$3^3 \times 3^2 = \square^{\square}$

11 $4^2 \times 4^4 =$ ______

$4^2 \times 4^4 = \square^{\square}$

Now algebraic indices - same process

12 $x^2 \times x^3 =$ ______

$x^2 \times x^3 = \square^{\square}$

13 $a \times a^3 =$ ______

$a \times a^3 = \square^{\square}$

You can multiply indices by adding the indexes. Multiply these, leave answer in index form

Example

$5^2 \times 5^4 = 5^{2+4} = 5^6$

14 $7^4 \times 7^2 = \square^{\square+\square} = \square^{\square}$

15 $9^3 \times 9^5 = \square^{\square+\square} = \square^{\square}$

No power - it's like there is a '1' there

16 $4^6 \times 4 = \square^{\square+\square} = \square^{\square}$

17 $5 \times 5^2 = \square^{\square+\square} = \square^{\square}$

18 $7 \times 7 = \square^{\square+\square} = \square^{\square}$

19 $3^9 \times 3^7 = \square^{\square+\square} = \square^{\square}$

Now algebraic indices - same process

20 $x^3 \times x^3 = \square^{\square+\square} = \square^{\square}$

21 $n^5 \times n = \square^{\square+\square} = \square^{\square}$

22 $a \times a^3 = \square^{\square+\square} = \square^{\square}$

23 $h^8 \times h^4 = \square^{\square+\square} = \square^{\square}$

Forget the middle step, answer only. Triples are just the same.

24 $5^8 \times 5^4 = \square^{\square}$

25 $7^3 \times 7^9 = \square^{\square}$

26 $7^3 \times 7^9 \times 7^2 = \square^{\square}$

27 $3 \times 3^6 \times 3^3 = \square^{\square}$

28 $e^4 \times e^3 \times e^5 = \square^{\square}$

29 $k \times k^5 \times k = \square^{\square}$

30 $h^2 \times h \times h^7 = \square^{\square}$

We now have unlike terms. At last, maths questions that have two correct answers! Simplify.

Example

$a^2 \times b^2 \times a^4 \times b = a^6b^3$ *or* b^3a^6

31 $a^6 \times b =$ ______ *or* ______

32 $c^4 \times b^5 =$ ______ *or* ______

33 $k^9 \times t^3 =$ ______ *or* ______

Just one answer from now on

34 $x^3 \times a^5 \times x^2 \times a =$ ______

35 $y^2 \times c^3 \times y^4 \times y^5 =$ ______

36 $a^5 \times b^2 \times a \times b =$ ______

37 $d \times d^2 \times c \times d^5 =$ ______

Less '×' signs - but the same process

38 $bc \times b^2 =$ ______ *or* ______

39 $ay \times y^3 =$ ______ *or* ______

40 $g^3k \times k^3g^2 =$ ______

41 $ab \times ab =$ ______

42 $abc \times abc =$ ______

43 $b^2ac^3 \times acb =$ ______

44 $c^4a^2b^3 \times b^4c^3a^6 =$ ______

45 $ab^3c \times a^2b \times ab^2 =$ ______

Which of these is true?

46 $m^3 \times m^2 =$ ○ m^6 ○ m^5

47 $e^6 \times e =$ ○ e^6 ○ e^7 ○ e^0 ○ e

48 $a^2 \times a^5b =$ ○ $a^{10}b$ ○ a^7b^5 ○ a^2b^5 ○ a^7b